

Compressed Christoffel Width and Torsion-Free Primitive-Power Elimination

Exact rank-two terminals, parallel contraction bounds,
and a costed route toward quasi-polynomial unknot recognition

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Abstract

We develop a compressed algebraic kernel for the group-certificate branch of unknot recognition. A cyclically reduced word in a rank-two free group is a power of a primitive element precisely when it is sign-coherent and its primitive-slope prefix heights have the minimum possible oscillation. A binary straight-line program can evaluate this oscillation by two bottom-up passes: no word expansion, Whitehead descent, compressed string equality, or root extraction is required. With S grammar nodes and B bits for the maximum expanded length, the query and certificate replay take $O(S(B + \log(S + 2)) + B^2)$ bit operations under standard arithmetic bounds.

For a certified classical knot-group presentation, a primitive-power relation permits killing its primitive root by torsion-freeness. In rank two this is an unknot certificate, even when additional relators remain and the generators are not meridians. In higher rank, a two-generator-supported relation permits an explicit projection $a \mapsto t^v$, $b \mapsto t^{-u}$. Disjoint projections can be batched. If unselected relators are kept as raw substitution circuits, a d -round contraction schedule has grammar size at most $S_0 + O(r_0 B_0 2^d)$ and can be discovered greedily or checked along a supplied schedule in $\text{poly}(S_0, r_0, m, B_0) 2^{O(d)}$ time. The schedule must actually succeed; no universal depth or producer bound is asserted.

We also give a complete compressed terminal for certified rank-two one-relator presentations, a normal-consequence search bound, and an integration plan. The accompanying implementation passes exhaustive comparison with literal Whitehead reduction on 29,540 cyclically reduced words, independent replay, small closed-braid/Jones cross-checks, and large nonuniform compressed tests. The contribution is the explicit arithmetic kernel and its costed composition with knot-group certificates, not a claim of first proving rank-two compressed primitivity or of completing unrestricted quasi-polynomial recognition.

Status. The local theorems below are proved with their stated hypotheses. The global quasi-polynomial implication is conditional on a controlled producer or a successful bounded-depth contraction run. The code is a tested research package, not an installed modification of the maintained recognizer and not a Lean formalization.

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1 Research objective and relation to the current repository

The intended input to the eventual recognizer is a classical knot diagram, not an arbitrary finite presentation with an unverified label saying that it is a knot group. The central objective remains a uniform bound of the form

$$T(n) \leq \exp(O((\log(n+2))^c))$$

in the diagram size n , for a fixed constant c . A fast algebraic subroutine only contributes to this objective when the cost of reaching its input, the size of that input, and the correctness of its connection to the original knot are all included.

The inspected repository has a certificate-oriented architecture. Its `group_certificates.tex` describes reconstruction of the complete Wirtinger presentation, exact defining-generator eliminations, and Whitehead moves. Its `relator_powers.tex` develops quotient-sized donor deletions and a pure-letter-power Euclidean kernel. The recent intake synthesis treats a polynomial compressed terminal for two *certified meridian* generators. The maintained `compressed_search.py` returns an internal search success only to a wrapper that rebuilds and checks the trace. These are useful constraints, rather than details to bypass [10, 11, 12, 14].

The present contribution is different in three respects. First, the rank-two test is a constant-dimensional arithmetic summary of a cyclically reduced grammar. Second, primitive *powers*, rather than just primitive relators or pure letter powers, supply an elimination rule in a torsion-free group. Third, a normalization-free parallel composition gives an explicit round-depth parameter that controls grammar growth. The terminal does not require a meridian marking on the current free generators. It does still require the stronger, indispensable fact that the *whole current presentation* is the group of the original knot.

1.1 What is classical, and what is being added

The classification of primitive conjugacy classes in F_2 by coprime lattice vectors and Christoffel words is classical. We use the formulation in Kassel–Reutenauer, Theorem 3.2 and Corollary 3.3 [1]; the earlier work of Osborne–Zieschang is another standard source [2]. Prefix ranges of a weighted word also form a familiar concatenation monoid. Neither of these facts is claimed as a new foundational discovery.

Kapovich’s July 2026 preprint proves deterministic polynomial-time compressed primitivity in rank two and describes a Euclidean automorphism construction followed by one compressed cyclic-reduction test [3, Section 8]. Consequently, “compressed rank-two primitivity is in $\mathbf{P}$ ” is not a new complexity-class claim here. Our kernel instead exploits the useful *already cyclically reduced* interface: after counts and one gcd, it uses only prefix extrema. It also recognizes all powers of primitive words and feeds a projection rule for entire presentations. This removes a final general-purpose reduction from this particular entry interface; it does not prove a uniformly better bound when arbitrary normalization is included.

The round-depth bound, the distinction between raw circuits and repeated normalization, and the independent certificate implementation are the main project-facing results. Priority beyond the sources examined has not been established. They should be integrated as proved deductions with explicit hypotheses, not advertised as a resolution of the full recognition problem.

1.2 Current complexity claims are not interchangeable

Lackenby's July 2026 manuscript gives an unknot-recognition algorithm through incompressible surfaces and hierarchies. Section 9 explicitly separates the number of visits to stages from running-time estimates, because some stages are not specified sufficiently precisely for the latter [9, Section 9]. This report neither replaces that geometric program nor interprets an algorithmic announcement as a bound for the repository implementation. Likewise, a positive certificate kernel, a complete restricted terminal, and a complete recognizer have different interfaces and are labeled separately below.

2 Input model and algebraic conventions

Let $F_r = F(x_1, \dots, x_r)$ be the free group on a fixed current basis. A word is freely reduced if no adjacent letters are mutual inverses. A nonempty freely reduced word is cyclically reduced if its first and last letters are not inverses. We include the empty word as cyclically reduced, but never as a nonzero power of a primitive element. An element is primitive if it belongs to a free basis. A *primitive-power word* represents $gu^d g^{-1}$, where u is primitive and d is a nonzero integer. Its inverse can absorb the sign of d , so certificates below use $d > 0$.

A binary SLP is a topologically ordered directed acyclic grammar. Node zero is empty; each later node is either a signed letter or a concatenation of two earlier nodes. Multiple root nodes may share subgrammars. Write S for the number of supplied nodes, including the empty node; m for the number of relator roots; L for the maximum expanded length of a supplied node, with $L \geq 1$ by convention; and

$$B = \max\{1, \lceil \log_2(L + 1) \rceil\}.$$

This maximum is important. If unreachable nodes are supplied and validated, their lengths are charged as well. Alternatively, prune to the ancestors of the roots first, retaining the cost of reading and remapping the input. For binary grammars $B = O(S)$, but L may be exponential in S .

The encoded input also contains node identifiers of $O(\log(S + 2))$ bits. We count additions and comparisons of B -bit integers in $O(B)$ time. Classical gcd, exact division, and multiplication may conservatively be charged $O(B^2)$ each. The classical quadratic bound for an entire Euclidean gcd computation is used, not the product of a pessimistic quadratic division bound with a linear number of divisions. All subsequent statements remain valid with any other fixed polynomial arithmetic bounds.

For $F(a, b)$ write

$$\text{ab}(W) = (\sigma_a(W), \sigma_b(W)).$$

A word is *sign-coherent* when neither generator appears with both signs. After independently inverting a and b , every nonempty sign-coherent word is positive. Counts of absent generators are zero; their sign choice is irrelevant. All word-level negative answers assume that the root is actually cyclically reduced. Invalid normalization is a precondition failure, not a non-primitivity answer.

3 Minimum Christoffel width

We first prove the arithmetic test, including its power version. Let W be a positive word with P letters a and Q letters b , not both zero. Put

$$d = \gcd(P, Q), \quad p = P/d, \quad q = Q/d, \quad \ell = p + q.$$

For the prefix of length j , let A_j, B_j count its a and b letters and set

$$H_j = qA_j - pB_j, \quad \Omega(W) = \max_{0 \leq j \leq P+Q} H_j - \min_{0 \leq j \leq P+Q} H_j. \quad (1)$$

The empty prefix and the full prefix are included. In particular, $H_0 = H_{P+Q} = 0$.

Lemma 3.1 (The unavoidable width). *If $p, q > 0$, then $\Omega(W) \geq p + q - 1$.*

Proof. Every a step adds q and every b step adds $-p$. Modulo $\ell = p + q$ these increments are equal. Thus $H_j \equiv jq \pmod{\ell}$. Coprimality of p, q implies $\gcd(q, \ell) = 1$, so the first ℓ prefix heights represent all ℓ residues. An interval of integer width less than $\ell - 1$ contains fewer than ℓ integers. It cannot contain these heights. $\square$

Lemma 3.2 (Equality forces a periodic rotation path). *Suppose $p, q > 0$ and $\Omega(W) = \ell - 1$. Then W is a repetition, d times, of a cyclic conjugate of a Christoffel word with counts (p, q) .*

Proof. Let $h = \min H_j$. Every height lies in $I = \{h, h + 1, \dots, h + \ell - 1\}$. The possible next heights from an integer $z \in I$ are $z + q$ and $z - p$, which differ by exactly ℓ . Exactly one is in I : the first is in I precisely when $z \leq h + p - 1$, and the second precisely when $z \geq h + p$. Therefore the next letter is determined by the current height. This is the rotation $z - h \mapsto z - h + q \pmod{\ell}$, realized on consecutive integer representatives. It visits all heights and returns after exactly ℓ steps. Its letters consist of p occurrences of a and q of b . Starting at a different point merely cyclically rotates the same word. Since $|W| = d\ell$, W follows this cycle exactly d times. The rotation word is the usual lower Christoffel word, up to the conventional choice of initial point. $\square$

Lemma 3.3 (Christoffel roots are primitive). *For coprime $p, q \geq 0$, the rotation word $C(p, q)$ starting at height zero is primitive in $F(a, b)$.*

Proof. The boundary cases are $C(1, 0) = a$ and $C(0, 1) = b$. For $p \geq q > 0$ write $p = kq + r$. The rotation path decomposes into b steps preceded by k or $k + 1$ occurrences of a , and gives

$$C(p, q) = \phi_k(C(r, q)), \quad \phi_k(a) = a, \quad \phi_k(b) = a^k b. \quad (2)$$

One can verify the block description directly from the rule in Lemma 3.2: before a b step the height must reach at least p , and removing k a steps per b leaves the rule for (r, q) . Similarly, when $q = kp + r > p$,

$$C(p, q) = \psi_k(C(p, r)), \quad \psi_k(a) = ab^k, \quad \psi_k(b) = b. \quad (3)$$

Both maps are Nielsen automorphisms, with the indicated powers negated in their inverses. Euclidean descent reaches one of the boundary cases, so $C(p, q)$ is an automorphic image of a generator. These are also the standard Christoffel morphisms; compare [1, Section 3]. $\square$

Theorem 3.4 (Exact primitive-power width criterion). *Let W be a nonempty cyclically reduced word in $F(a, b)$. The following are equivalent:*

- (i) W is a conjugate of a nonzero power of a primitive element;
- (ii) W is sign-coherent and, after its sign normalization,

$$\Omega(W) = p + q - 1.$$

When these conditions hold, the primitive-power exponent is $d = \gcd(P, Q)$. In particular, W itself is primitive exactly when $d = 1$. The criterion includes pure one-letter words, where $\ell = 1$ and the width is zero.

Proof. For a positive word with both letters, equality implies the assertion by Lemmas 3.2 and 3.3. The cyclic root has counts (p, q) , so it cannot be a further proper power: any such power would divide both p and q . This proves the exponent assertion.

For the converse, cyclically reduce a primitive root. The classical rank-two classification says that this root is a cyclic conjugate of a Christoffel word after independently choosing the signs of the generators [1, Corollary 3.3]. Since the root is cyclically reduced, its positive powers have no cancellations at their joins. A cyclically reduced conjugate is a cyclic rotation. It is therefore sign-coherent and traverses the same ℓ -height rotation cycle repeatedly, with width $\ell - 1$. For a single occurring generator the reduced word is a nonzero power of that generator, and the stated zero-width computation is immediate. Finally, a proper power cannot be primitive because its abelianization vector is not primitive. $\square$

3.1 Signed implementation formula

Instead of changing the grammar, compute the signed exponent vector (U, V) and $d = \gcd(|U|, |V|)$. For a nonempty coherent root $d > 0$. Put $(u, v) = (U/d, V/d)$ and give the original signed letters the weights

$$a \mapsto v, \quad a^{-1} \mapsto -v, \quad b \mapsto -u, \quad b^{-1} \mapsto u. \quad (4)$$

These heights are either the normalized heights or their negatives, so oscillation is unchanged. The acceptance threshold is $|u| + |v| - 1$. This avoids changing leaf labels and deals uniformly with negative slopes.

3.2 A necessary safeguard against abelianization shortcuts

For coprime $p, q > 1$, the positive block word $a^p b^q$ has height range $[0, pq]$. Its excess over the minimum is exactly

$$\Omega(a^p b^q) - (p + q - 1) = (p - 1)(q - 1) > 0. \quad (5)$$

Thus coprime exponent sums emphatically do not imply primitivity. The presentation $\langle a, b \mid a^2 b^{-3} \rangle$, for example, is the standard trefoil presentation, and its normalized word has width 6 rather than the required 4. The arithmetic test distinguishes order information that the exponent vector loses.

4 Two-pass compressed queries and independent replay

For each grammar node store its four signed occurrence counts, its first and last letters, its length, and a freely reduced flag. At a concatenation XY , counts and lengths add. Endpoints are inherited from the nonempty children. The reduced flag is true exactly when both child flags are true and either a child is empty or the last letter of X is not the inverse of the first letter of Y . A root

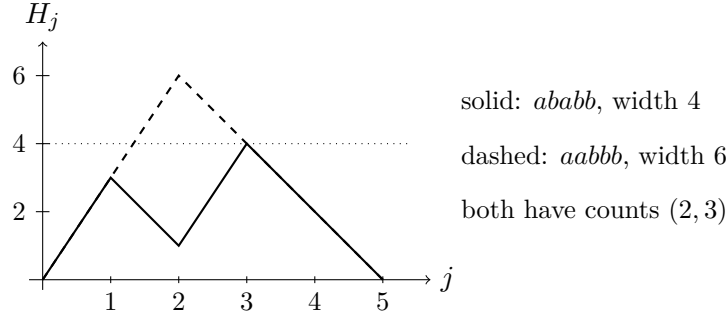


Figure 1: Minimum-width Christoffel order versus the same coprime counts in nonprimitive block order. The comparison is exact over the integers.

is cyclically reduced exactly when this flag is true and its nonempty endpoints are not inverses. Induction on the grammar proves all these statements without expansion.

Once the slope is fixed, define the profile of a word by

$$\mathcal{P}(X) = (t_X, m_X, M_X),$$

where t_X is its total weight and m_X, M_X are its minimum and maximum prefix weights, including the empty prefix. A letter of weight w has profile $(w, \min(0, w), \max(0, w))$. The empty word has profile $(0, 0, 0)$.

Lemma 4.1 (Prefix-profile product). *For any two words, with no reduction performed,*

$$(t, m, M) \star (s, n, N) = (t + s, \min(m, t + n), \max(M, t + N)) \quad (6)$$

is the profile of their concatenation. This product is associative.

Proof. Every prefix of XY is either a prefix of X or all of X followed by a prefix of Y . Their weights form the union of the prefix-weight set of X and its translate by t_X of the prefix-weight set of Y . This proves the formula. Applying the same description to three successive prefix sets proves associativity, or it follows from associativity of concatenation and the explicit formulas for extrema. $\square$

Theorem 4.2 (Compressed query bound). *On a validated binary SLP with cyclically reduced root, primitive-power recognition and extraction of (u, v, d, Ω) take*

$$O(S(B + \log(S + 2)) + B^2) \quad (7)$$

bit operations and $O(SB + S \log(S + 2))$ bits of working storage. A certificate consisting of the root identifier and these four integers can be independently checked within the same asymptotic bound. The bound becomes $O(S^2)$ for binary grammars, up to harmless small-input conventions.

Proof. The count/endpoint pass has constant-dimensional integer summaries. Counts are at most L . The gcd and exact divisions have the stated quadratic bit cost. Each letter weight has $O(B)$ bits and every prefix height has absolute value at most L^2 , hence at most $2B + O(1)$ bits. Formula (6) uses only a fixed number of additions and comparisons per node. Reading and validating pointers contributes $O(S \log(S + 2))$.

For replay, derive all counts and endpoints afresh from the original grammar, check $d > 0$, $\gcd(|u|, |v|) = 1$, $\text{ab}(W) = d(u, v)$, sign coherence and cyclic reduction, and recompute the profile using weights (4). Finally check $\Omega = |u| + |v| - 1$. The verifier need not trust any producer summary. Storing total weights per node avoids multiplying a slope by a count vector at every internal node; only a constant number of final integer products are needed. The same bound follows. Since $B = O(S)$, both passes are quadratic in the binary grammar-node parameter in the worst case. $\square$

4.1 What is deliberately not inside this bound

The root must already be cyclically reduced. General compressed free and cyclic reduction are polynomial-time operations, by Lohrey’s results as exposed in Schleimer [4, Theorem 3.3 and Corollary 3.6]. If normalization maps an input of size S to an output of size S' , the correct combined bound is

$$T_{\text{normalization}}(S) + O(S'(B' + \log(S' + 2)) + (B')^2),$$

not automatically the right side of (7) with the old S . The standalone reference kernel rejects an unreduced input; it does not pretend to implement this general normalization theorem. The inspected repository already has a separate exact compressed-word backend [13].

The code uses no random fingerprints, approximate equality, floating-point heights, or expanded-length loops in the kernel. Its resource limits are interrupts. Exceeding a node, bit-length, or external cancellation budget cannot return a mathematical negative answer.

5 Shared scans over all relators

For a genuine two-generator knot-group presentation, the integer exponent matrix has row lattice of rank one: its cokernel is the knot abelianization $\mathbb{Z}$. Thus all nonzero exponent rows lie on one rational line. From one nonzero row compute a primitive direction (u, v) , choosing its sign canonically. Collinearity of another row (A, B) is exactly

$$vA - uB = 0.$$

A single profile pass with weights $(v, -u)$ then supplies the widths of *all* relator roots in the shared grammar.

Proposition 5.1 (Shared-slope scan). *For m cyclically reduced roots with collinear exponent vectors, one profile pass identifies every primitive-power root. A conservative bit bound is*

$$O(S(B + \log(S + 2)) + m(B^2 + \log(S + 2))). \quad (8)$$

Proof. Use the common direction for the profile pass. For each nonempty coherent root with exponent vector $(A, B) \neq (0, 0)$, let $d = \gcd(|A|, |B|)$. Its own primitive direction is either (u, v) or its negative. Negating weights leaves width unchanged, so the common threshold $|u| + |v| - 1$ applies. A nonempty root with zero exponent vector cannot be a primitive power. Theorem 3.4 therefore decides every root. Collinearity tests, per-root gcds, and exact arithmetic can be charged $O(mB^2)$; they must not be described as unit-cost merely because the profile itself uses only additions. $\square$

This is a reuse optimization, not an inference of topology from Smith normal form. Collinearity alone does not even show that the abelianization is $\mathbb{Z}$: the gcd of the scalar row multiples must

also be one. More importantly, abelianization $\mathbb{Z}$ does not show that a presentation is a classical knot group. A verified geometric origin and an equivalence trace are still needed before invoking torsion-freeness or the unknot theorem.

6 From primitive powers to whole-presentation elimination

The following topological facts form a small, explicit interface to the algebra. A classical knot exterior is irreducible, its first homology is $\mathbb{Z}$, and its fundamental group is torsion-free. For torsion-freeness, one may use asphericity together with the fact that a group admitting a finite-dimensional classifying space has no nontrivial finite cyclic subgroup; see [5, (C.1)–(C.2)]. The classical Dehn/Loop Theorem implies that a knot group is infinite cyclic precisely for the unknot [6].

For completeness, the latter implication can be seen directly at the boundary torus. If $\pi_1(E_K) = \mathbb{Z}$, the inclusion-induced map $\pi_1(\partial E_K) = \mathbb{Z}^2 \rightarrow \pi_1(E_K)$ cannot be injective. The Loop Theorem gives a compressing disk. An irreducible orientable knot exterior with compressible torus boundary is a solid torus: compressing the boundary gives a sphere, which bounds a ball, and reversing the compression attaches a single one-handle. The knot is therefore the unknot. These facts apply to classical knots in S^3 , not to arbitrary links, virtual diagrams, or presentations with only a matching abelianization.

Theorem 6.1 (Torsion-free root elimination). *Suppose a presentation $\langle X \mid \mathcal{R} \rangle$ presents a torsion-free group G . If an actual relator or a verified normal consequence is a conjugate of w^d , with $d \neq 0$ and w primitive in $F(X)$, then adding $w = 1$ preserves G . In a basis containing w , one may eliminate its generator and project all remaining relators.*

Proof. The represented element satisfies $w^d = 1$ in G . Torsion-freeness implies $w = 1$ already in G , so adjoining that relation makes no further quotient. Primitivity supplies a basis with first member w . The added relation kills that generator, and the usual defining-generator elimination gives the claimed presentation on one fewer generator. All other relators are substituted rather than discarded. $\square$

Remark 6.2 (A root rule is not an unrestricted Tietze rule). The presentation $\langle a, b \mid a^2 \rangle$ defines $C_2 * \mathbb{Z}$. Replacing a^2 by a changes it to $\mathbb{Z}$. Thus a syntactic primitive-power certificate is not by itself a proof that root extraction preserves an arbitrary presented group. When $d = 1$, the extra torsion-free hypothesis is unnecessary; for $d > 1$ it must be justified by the caller's verified context.

6.1 An explicit two-letter projection without basis expansion

Theorem 6.3 (Fused primitive-pair projection). *Let w be primitive in the free factor $F(a, b) \leq F(X)$, with $\text{ab}(w) = (u, v)$ and $\gcd(|u|, |v|) = 1$. Let t be a new generator and define*

$$\pi(a) = t^v, \quad \pi(b) = t^{-u}, \quad \pi(x) = x \quad (x \in X \setminus \{a, b\}). \quad (9)$$

Then

$$F(X)/\langle\langle w \rangle\rangle \cong F((X \setminus \{a, b\}) \cup \{t\})$$

via π . Consequently, under Theorem 6.1, a two-letter primitive-power relation permits replacing every relator by its image under (9). There is no need to expand a Nielsen basis or a conjugating word.

Proof. Because w is primitive, $F(a, b)/\langle\langle w \rangle\rangle$ is infinite cyclic. The character $a \mapsto v, b \mapsto -u$ annihilates w and is surjective because $\gcd(|u|, |v|) = 1$. Hence the induced epimorphism from this infinite cyclic quotient to $\mathbb{Z}$ is an isomorphism, and its kernel in $F(a, b)$ is exactly $\langle\langle w \rangle\rangle$. Passing to the free product with the remaining generators proves the assertion. More explicitly, Bezout coefficients α, β satisfying $v\alpha - u\beta = 1$ give the inverse generator $t = \overline{a^\alpha b^\beta}$ in the quotient. No commutativity is imposed on the other free generators. $\square$

The root detected by Theorem 3.4 is only specified up to conjugacy and inversion. Neither ambiguity changes its normal closure. A primitive of a free factor $F(a, b)$ remains primitive in $F(X)$ by extending its basis with the other generators. These observations discharge both potential ambiguities in using the local classifier inside a larger presentation.

Corollary 6.4 (Two arbitrary generators suffice at the positive terminal). *Let $\langle a, b \mid R_1, \dots, R_m \rangle$ be certified to present the group of a classical knot. If any actual relator or verified normal consequence is a nonzero power of a primitive element of $F(a, b)$, then the knot is the unknot.*

Proof. Root elimination makes the presented group a quotient of an infinite cyclic group. Since its abelianization is $\mathbb{Z}$, this quotient is $\mathbb{Z}$ itself. The knot-group criterion above finishes the proof. No remaining relation needs to be assumed redundant, and neither a nor b needs to be a meridian. $\square$

Failure to find such a relation in a supplied multi-relator list is only a failure of that positive terminal. It is not a nontrivial-knot certificate. An unknown relation in the normal closure may still have the required form.

6.2 An exact family without any torsion-free hypothesis

Theorem 6.5 (Aligned primitive-power presentations). *Suppose every nonempty relator of a rank-two presentation is a primitive power and all their primitive exponent vectors lie on the same line. Let $d_1, \dots, d_s > 0$ be their exponents and $g = \gcd(d_1, \dots, d_s)$. Then the presented group is $C_g * \mathbb{Z}$, with C_1 trivial. If all relators are empty, the group is F_2 instead.*

Proof. Primitive vectors on a fixed integral line differ only by sign. By the rank-two classification, the primitive roots are conjugate to a fixed primitive word w or to w^{-1} [1]. Thus their normal closure is $\langle\langle w^{d_1}, \dots, w^{d_s} \rangle\rangle$. Bezout's identity shows that w^g is a product of integer powers of these elements, and every w^{d_i} is a power of w^g . The normal closure is exactly $\langle\langle w^g \rangle\rangle$. Completing w to a basis produces $\langle w, z \mid w^g \rangle = C_g * \mathbb{Z}$. $\square$

For instance, take a nonuniform Christoffel word C and compressed exponents $N = 2^h, N^2 + 1$. The two-relator presentation

$$\langle a, b \mid C^N, C^{N^2+1} \rangle$$

is infinite cyclic. The shared width/gcd scan proves this without repeatedly subtracting a donor or comparing differently parsed copies of C . This extends the repository's pure-letter arithmetic example to a nonuniform primitive root and to independently rotated relators. It does not show that an arbitrary input knot has such a presentation.

Proposition 6.6 (A primitive anchor classifies arbitrary additional relators). *If one relator R is primitive with exponent vector (u, v) , then*

$$\langle a, b \mid R, W_1, \dots, W_k \rangle \cong \mathbb{Z}/h\mathbb{Z}, \quad h = \gcd_i |v\sigma_a(W_i) - u\sigma_b(W_i)|,$$

where $h = 0$ means $\mathbb{Z}$.

Proof. Quotient by R using Theorem 6.3. Each remaining relator becomes a power of t with the displayed exponent. Relations in a cyclic group generate the subgroup determined by their gcd. $\square$

This is an unconditional algebraic statement. It is useful for independently checking that no remaining relator was silently forgotten.

7 A complete rank-two one-relator terminal

Theorem 7.1 (One-relator cyclicity). *For a nonempty word $R \in F(a, b)$,*

$$\langle a, b \mid R \rangle \cong \mathbb{Z} \iff R \text{ is primitive in } F(a, b).$$

Hence, on a certified rank-two one-relator knot-group presentation, normalize R once and apply Theorem 3.4, accepting precisely when the primitive-power exponent is one. This is a complete decision at that interface, not merely a positive test.

Proof. The forward implication uses the Magnus normal-closure theorem: two elements of a free group with the same normal closure are conjugate or inverse-conjugate [7, 8]. Let the quotient map to $\mathbb{Z}$ send a, b to coprime integers A, B . A Nielsen change of basis transforms this primitive character to the coordinate projection $F(s, t) \rightarrow \mathbb{Z}$ with $s \mapsto 0, t \mapsto 1$. Its kernel is $\langle\langle s \rangle\rangle$. Translated back to the original basis, there is a primitive w such that $\langle\langle R \rangle\rangle = \ker(F_2 \rightarrow \mathbb{Z}) = \langle\langle w \rangle\rangle$. Magnus's theorem makes R conjugate to w or w^{-1} , hence primitive. Conversely, quotienting a free basis by one of its members leaves an infinite cyclic group. $\square$

If R cyclically reduces to the empty word, the presentation is F_2 and not cyclic. In a purported knot presentation this also fails the necessary abelianization test. The topology wrapper must distinguish invalid input provenance from a nontrivial-knot outcome.

It is essential not to extend the negative direction to an arbitrary multi-relator presentation. For example, $\langle a, b \mid (ab)^2, (ab)^3 \rangle \cong \mathbb{Z}$, although neither relator is primitive. This also illustrates why the torsion-free positive terminal and the aligned-family theorem are useful alongside the one-relator test.

7.1 A justified deficiency-one starting presentation for braids

A closed r -strand braid has the Artin presentation

$$\langle x_1, \dots, x_r \mid \beta(x_i)x_i^{-1}, 1 \leq i \leq r \rangle.$$

Each local Artin automorphism preserves $x_1 \cdots x_r$. If the first $r - 1$ relations hold, the equality $\beta(x_1) \cdots \beta(x_r) = x_1 \cdots x_r$ implies the last relation. Thus the last relator can be removed with this explicit justification. Starting with the resulting $r - 1$ relators, each defining-generator elimination removes one generator and one defining relation. At rank two, there is one remaining designated relation, possibly accompanied by empty slots. The endpoint criterion above therefore applies whenever such a certified reduction reaches it.

The supplied braid demonstration conservatively retains all Artin relators and implements only the positive terminal. The deficiency-one deletion and the complete negative endpoint are proved here but are not represented as already installed production features.

8 Parallel contractions with an explicit grammar-growth bound

The next result addresses a different bottleneck: composing many safe local reductions without repeatedly paying an uncontrolled normalization cost. The crucial restriction is stated as part of the algorithm, not left implicit.

Definition 8.1 (Normalization-free contraction round). At the start of a round the presentation is represented by a shared binary grammar, whose relators may be unreduced. Select some current relator roots whose *spelled words* are cyclically reduced primitive powers supported on pairwise disjoint pairs of current generators. Verify their local certificates. Apply all maps (9) simultaneously, and retain all other relators as raw substitution circuits. A selected donor may be replaced by an empty root after the checked projection, since its image is freely trivial by Theorem 6.3. Its old slot and proof record remain in the trace. No free or cyclic reduction is run on the other roots.

Disjointness is a syntactic condition on generator support. It makes the maps independent and allows all new image powers to be built before one pass through the old grammar. It is not a claim that arbitrary overlapping primitive roots commute as elimination operations.

Theorem 8.2 (Normalization-free round-depth bound). *Let an initial torsion-free presentation have r_0 generators, m relator slots, S_0 grammar nodes, and maximum length-bit bound B_0 . For a run of d nonempty normalization-free contraction rounds using current relators, there is an implementation satisfying*

$$B_j \leq 2^j B_0, \quad (10)$$

$$S_j \leq S_0 + Cr_0 B_0 (2^j - 1) \quad (11)$$

for an absolute constant C , after pruning or remapping unreachable nodes. The run can be checked along a supplied schedule, or produced by the explicit greedy disjoint-pair scan until it stalls, in

$$\text{poly}(S_0, r_0, m, B_0) 2^{O(d)} \quad (12)$$

bit operations. Here a “run of d rounds” means the rounds that actually occur; the theorem does not say that the run reaches one or two generators.

Proof. Let $L_j \geq 1$ be the maximum expanded node length at round j . For every accepted pair, the primitive coordinates u, v have absolute values at most the length of its relator, hence at most L_j . Each original letter is therefore replaced by a power of length at most L_j , or by a single unaffected letter. Every old node expands to at most L_j^2 letters. New power nodes have length at most L_j . Thus $L_{j+1} \leq L_j^2$, with the convention $L_j \geq 1$, and $\lceil \log_2(L_j^2 + 1) \rceil \leq 2B_j$. This proves (10) by induction.

Binary powering constructs the positive and negative images for a generator pair with $O(B_j)$ new nodes. There are at most $r_0/2$ pairs. Copy each reachable old concatenation at most once, redirecting leaves to their image roots; shared images and shared old subexpressions are not expanded. This gives

$$S_{j+1} \leq S_j + Cr_0 B_j.$$

Summing this recurrence and (10) proves (11). Empty donor slots need no extra nonempty nodes.

To find eligible relators, compute endpoints, reduced flags, and supports. It is enough to remember at most three distinct support labels: three already makes a root ineligible for this two-letter kernel.

For each eligible two-letter root, extract and rename its ancestor grammar and apply Theorem 4.2. Even a separate scan for every relator costs a fixed polynomial in S_j, m, r_0, B_j . A greedy disjoint matching is another polynomial operation; no enumeration of all matchings is required for this stated run. Similarly a supplied matching can be checked in polynomial time. Substitution and node remapping have polynomial bit cost. There are at most $r_0 - 1$ nonempty rank-reducing rounds. Substituting (10)–(11) into a fixed polynomial and summing yields (12).

Finally, the group stays unchanged by Theorems 6.1 and 6.3. Each simultaneous set of killed primitive roots is already trivial in the current torsion-free group. The quotient of the free group by their normal closure is the free product of the corresponding rank-one factors and the unaffected generators. Projecting every remaining relator gives the same presented group. Torsion-freeness is thus available again in the next round. $\square$

Corollary 8.3 (A quantitative restricted route). *Suppose an n -size knot input is converted, with a verified trace and cost $V(n)$, to a presentation with $S_0, r_0, m, B_0 \leq n^{O(1)}$. If a produced normalization-free run reaches a valid cyclic terminal in d rounds, its additional work is $n^{O(1)}2^{O(d)}$. In particular, $d = O(\log n)$ gives polynomial additional work, and $d = O((\log n)^2)$ gives $n^{O(\log n)}$ additional work.*

A complete restricted recognizer results if the producer is guaranteed to reach a rank-two one-relator endpoint on every input in the promised class, with the stated bounds, and its final normalization and empty-relator checks are performed once by a fixed polynomial compressed-word algorithm.

Proof. Apply Theorem 8.2. At a one-generator endpoint, each raw relator reduces to a power determined by its exponent sum, so cyclicity can be checked without expanded reduction. At a rank-two one-relator endpoint, use Theorem 7.1. A fixed polynomial final normalization on a grammar of size $n^{O(1)}2^{O(d)}$ stays within that form. The cost $V(n)$ is added, not suppressed. $\square$

8.1 Why not normalize every round?

Suppose all that is known about a normalization routine is that it maps a grammar of size S to one of size at most S^c . Calling it after every round permits a recurrence $S_{j+1} \leq (S_j + Cr_0B_j)^c$. Iteration can produce a bound with an exponent depending exponentially on j . Merely calling each individual step “polynomial” does not give (12). The raw-circuit restriction avoids this issue. It may make the search stall sooner because an algebraically useful but unreduced relator is ineligible. That loss of applicability is explicit and is the subject of a research question, not something concealed by a free normalization oracle.

8.2 What the depth parameter does not prove

A small maximal matching in one round need not lead to a favorable later presentation. Even if some sequence of choices reaches a good endpoint, a particular greedy schedule might not. The theorem bounds the work of a specified successful run, or of the particular greedy run that is actually produced; it does not turn an existential good schedule into a polynomial search for one. A search over schedules must have its own branching bound.

Likewise, the theorem uses *current relators*. Replacing that requirement by “some short-looking normal consequence exists” introduces a proof-search cost. The next section records one honest way to charge that cost.

9 A canonical normal-consequence target and its search cost

The preceding terminal accepts a suitable relation once it is present. There is also a useful way to specify exactly what an unrestricted positive search would need to prove at a certified rank-two interface.

Theorem 9.1 (A fixed primitive target). *Suppose $P = \langle a, b \mid R_1, \dots, R_m \rangle$ has abelianization $\mathbb{Z}$. Let the lattice generated by the exponent vectors of the relators be $\mathbb{Z}(u, v)$, with $\gcd(|u|, |v|) = 1$. Choose the signed Christoffel word C with exponent vector (u, v) . Then*

$$P \cong \mathbb{Z} \iff C \in \langle\langle R_1, \dots, R_m \rangle\rangle \subset F(a, b).$$

In particular, for a certified classical knot-group presentation, this normal-closure membership condition is equivalent to unknottedness.

Proof. Every relator has zero image under the primitive character $\chi(a) = v, \chi(b) = -u$. The kernel of this character is $\langle\langle C \rangle\rangle$, by Theorem 6.3. Thus $N = \langle\langle R_1, \dots, R_m \rangle\rangle \subseteq \langle\langle C \rangle\rangle$. If $C \in N$, equality holds and the quotient is $\mathbb{Z}$. Conversely, if $F_2/N \cong \mathbb{Z}$, the map induced by χ is an epimorphism from $\mathbb{Z}$ to $\mathbb{Z}$, hence an isomorphism. Its kernel is zero, so $N = \ker \chi = \langle\langle C \rangle\rangle$. The final assertion uses the knot-group criterion. $\square$

The lattice hypothesis is more than pairwise collinearity: the greatest common divisor of the integer multiples on the primitive line must be one. It is supplied automatically by a correct knot-group provenance chain, or can be checked arithmetically as a necessary consistency condition. It is not itself a proof that the presentation comes from a knot.

The Euclidean construction of Lemma 3.3 builds C as a small grammar. A conservative $O(B^2)$ node bound follows from $O(B)$ Euclidean stages, each using binary powers of B -bit quotients; a more careful sum of quotient bit lengths gives $O(B)$ nodes. Either bound is polynomial. A small target is therefore available without solving the membership problem. This distinction prevents an important circular argument: the exponent vector identifies the target, but it does not prove that the target is a consequence of the original relations.

Definition 9.2 (Normal-derivation program). A program starts with the m input relators, regarded as identities in the presented group. A new instruction may invert an earlier identity, multiply two earlier identities, or conjugate an earlier identity by one signed basis letter. The program length is the number of new instructions. A chosen output is a *verified normal consequence* after its instruction list has been replayed.

Proposition 9.3 (A costed bounded search). *At a rank-two interface with initial grammar size S , all normal-derivation programs of length at most k can be searched for an output equal to the fixed target C in*

$$(m + k + 2)^{O(k+1)} \text{poly}(S + k + \text{bits}(C)) \quad (13)$$

bit time, where $\text{bits}(C)$ means its compressed encoding size. There is no expanded-word loop in this theoretical enumeration bound.

Proof. At an instruction there are at most $m + k$ earlier identities. An inversion chooses one, a product chooses two, and a conjugation chooses one and one of four signed letters. Including a choice of output, the number of programs is at most $(c(m + k + 2)^2)^{k+1}$ for an absolute constant c .

Store every identity and its inverse as raw circuits. Initially this at most doubles the input grammar. An inverse instruction swaps the two references; a product adds one concatenation and its reversed inverse; a one-letter conjugation adds a constant number of nodes. Therefore a program uses $O(S + k)$ nodes before testing its output. Append C^{-1} , normalize this one final word by a fixed polynomial compressed free-group algorithm, and test whether it is empty [4]. Intermediate identities are not normalized. Reading and replaying the instruction list, and testing the output, have the polynomial factor in (13). $\square$

If $S, m, \text{bits}(C) \leq n^{O(1)}$ and $k = O(\log n)$, this is $n^{O(\log n)}$. This is a genuine search bound under a quantified proof length, rather than a bound on checking a witness that an unspecified oracle supplies. It is not a bound for arbitrary knots: neither a uniformly small rank-two presentation nor the required logarithmic normal-derivation length has been proved here.

With no length cutoff the search is complete for the positive membership question. A member of the normal closure is a finite product of conjugates of input relators and their inverses; conjugation by a finite word can be spelled as successive signed-letter conjugations. This proves eventual success when the target belongs to the normal closure. It supplies no useful uniform time bound and no negative verdict after a failed bounded search. The enumeration theorem is not implemented in the accompanying package.

9.1 Why balanced existence alone is not the missing algorithm

There is an instructive abstract observation. If a presentation really presents $\mathbb{Z}$, write the images of its generators as integers $n_1, \dots, n_r$. For a pair with nonzero image vector (n_i, n_j) , put $g = \gcd(|n_i|, |n_j|)$ and choose a primitive word with vector $(n_j/g, -n_i/g)$. This word maps to zero, so is a normal consequence. The corresponding projection merges the pair into a generator whose image is g , up to sign. If both images are zero, a primitive letter is already an identity and can be eliminated. Thus, *if arbitrary true normal consequences are allowed at no search cost*, one can arrange a balanced rank contraction in logarithmically many levels.

That italicized qualification is the entire obstruction. The argument does not provide short derivations of those identities from the supplied relators, and it assumes the very cyclicity being decided. It is an existence observation about a known cyclic group, not a recognition algorithm and not a producer for Theorem 8.2. Proposition 9.3 is one precise place to charge the missing information.

10 Certificate architecture and implementation boundaries

The delivery separates a mathematical decision about a free word from a verdict about a knot. In particular, an API user cannot obtain a sound knot verdict by setting a Boolean `is_knot_group` on an arbitrary presentation. The hierarchy of interfaces is summarized in Table 1.

10.1 Small arithmetic evidence, independently reconstructed

The word certificate records its root identifier, signed primitive vector (u, v) , positive exponent d , and claimed width. The checker reconstructs counts, free-reduction flags, endpoints and all profiles from the supplied DAG. It rejects a false count, a nonprimitive slope, an incorrect exponent, a noncoherent root, invalid pointers and a nonminimal width. Boolean values are not accepted

Interface	Checked data	Meaning of success
Compressed word kernel	Grammar, reduction, signs, counts, exact height range	Primitive power in the specified free basis
Independent word replay	Original grammar and compact integer certificate	The same algebraic assertion, without producer helpers
Rank-two knot terminal	Complete input-to-presentation provenance plus word replay	Unknot, by torsion-freeness and abelianization
Pair projection prototype	Disjoint supports and local word certificates	Conditional torsion-free presentation contraction
Braid demonstration	Source braid, every initial relation, defining eliminations, terminal replay	Unknot of this particular braid closure

Table 1: These interfaces are intentionally not interchangeable.

in place of integer fields. Large certificate integers are serialized as hexadecimal strings to avoid language-runtime limits on very long decimal conversions.

The supplied `checker.py` does not import `christoffel.py`. Tests disable the producer helpers while replaying valid evidence. This is implementation independence, not a formal proof of the checker in a theorem prover. Both programs still rely on the host language’s exact integer arithmetic and on the mathematical argument in this report.

A malformed certificate is rejected. Cancellation, a bit-length ceiling or a work allowance being exhausted is a resource event and propagates out of the algebraic API. In a recognition portfolio it must become an inconclusive local attempt, or a globally cancelled query, never a negative topological answer. The demonstration modules have explicit caps but are not a substitute for the maintained recognizer’s complete resource-policy audit.

10.2 The read-only WordArena adapter

The inspected `WordArena` stores empty node zero, terminal rules (`'t'`, `signed_letter`) and binary rules (`'c'`, `left`, `right`) [13]. The adapter extracts the ancestors of selected roots, renames two current generators, and charges the supplied arena’s cooperative `tick` callback while remapping and querying. It returns algebraic evidence only. Its tests use a schema-compatible arena stub; they do not constitute a run of the full maintained repository suite.

There are two concrete integration traps in the inspected search [14]. The initial conversion calls free reduction, so a new kernel hook must not assume that every input root has already been *cyclically* reduced. Later paths do cyclically reduce their substituted roots. Second, the old success wrapper expects a trace ending with one surviving generator and no nonempty relators. Returning `True` from its internal search at a two-generator primitive-power terminal would violate that wrapper’s contract. A new versioned terminal record and an independent public replayer are required. The supplied integration note specifies this change but does not claim to install it.

10.3 A source-bound braid experiment

For a closed braid the demonstration constructs its Artin presentation literally, then tries defining-generator eliminations until two generators remain. The terminal is tested either with the compressed width kernel or with a literal Whitehead oracle. The verifier has separate implementations of the

Artin updates, substitution and cancellation, reconstructs all initial relations, and checks every elimination and the final word certificate against the original braid. The small demonstration is thus more than a free-standing claim about an invented presentation.

Nevertheless, its literal Artin words may grow exponentially, and its elimination choices may stall. It is an intentionally capped experiment, not a uniformly efficient braid-to-small-presentation producer. A failure returns `INCONCLUSIVE`. No negative knot decision is inferred from an absent primitive-power relator. The maintained pipeline is not modified by this experiment.

10.4 Parallel prototype scope

The prototype selects disjoint two-supported relators, verifies each local certificate with the independent word checker, applies the power projections, and retains unselected relators as raw circuits. It tests the syntactic eligibility and exact images used in Theorem 8.2. It does not independently certify that a supplied arbitrary presentation is torsion-free. Nor does the delivery include a second, fully independent whole-contraction trace checker. Its result is explicitly tagged as a *conditional torsion-free contraction*, not as a knot verdict.

The benchmark family used for this module is an abstract presentation family known to present $\mathbb{Z}$. It is not relabeled as a family of knot *diagrams*. This separation makes it possible to validate circuit handling and the depth parameter without falsely claiming a geometric producer.

11 Validation and controlled experiments

All measurements in this section are from completed runs of the delivered code. The raw JSON contains the environment, seeds and individual samples. The included table generator derives the displayed tables directly from those records. These measurements test the proposed kernel and research adapters; they are not a regression run or benchmark of the maintained `fastunknot` executable.

11.1 Independent mathematical cross-checks

The test runner executes 22 test methods with the following substantive coverage:

1. Every one of the 29,540 cyclically reduced words of lengths one through nine over $a^{\pm 1}, b^{\pm 1}$ is compared with a literal primitive-root and Whitehead oracle. Exactly 844 are accepted as primitive powers by both. The literal oracle uses explicit automorphism substitution, not the height criterion or a shared Whitehead cut formula.
2. There are 4,496 rotated, signed Christoffel-power checks and 1,200 random-parse checks. Counterexamples with coprime exponent vectors, malformed DAGs, unreduced words, cancellation, certificate mutations and exhausted budgets are tested separately.
3. A nonuniform grammar with 8,196 nodes and an expanded length of 6,940 binary digits is classified and independently replayed. Tests prohibit expansion on the large compressed cases. Equalities of a hash or a floating-point approximation are never used.
4. The braid experiment has an independent source-bound replayer. An additional 123 small, single-component random braid closures are checked against a separately implemented exponential

Kauffman-bracket cube; 90 receive positive certificates. Every certified one has normalized Jones polynomial one. This is a consistency check only: Jones identity is not used to certify the unknot.

5. Parallel-contraction tests reach initial rank 256, compare projected words with literal substitutions, enforce pair disjointness, preserve raw cancellation guards, and check the final rank-one relator exponent sums.

No unit-test count is presented as a proof of a universal theorem. The proofs and the tests discharge different obligations. The complete test log and machine-readable count ledger are retained in `data/`.

11.2 Paired compressed versus literal kernels

Each arm starts from the compact parameters and includes construction. The compressed arm includes independent certificate replay. The literal arm expands the word, extracts its literal root, and applies the independent Whitehead primitivity oracle. Five measured repetitions per arm are shuffled with a fixed seed. The reported ratios are quotients of medians, not confidence intervals.

Fib. steps	Power bits	Nodes	Length	Width (ms)	Literal (ms)	Ratio
8	2	12	110	0.049	0.396	8.0
12	2	16	754	0.071	2.334	32.9
16	2	20	5,168	0.101	14.743	146.3
10	5	18	2,448	0.076	1.570	20.7
10	9	22	37,008	0.105	11.721	111.3
10	13	26	589,968	0.156	181.438	1160.0

Table 2: Paired kernel medians, including construction and independent replay in the width arm. The baseline is literal, not the maintained compressed engine.

The largest displayed ratio is approximately 1.16×10^3 at expanded length 589,968. This is a genuine comparison with that *literal* baseline. It is not a speedup over the repository’s existing compressed Whitehead, compressed equality, or power-descent implementation. At larger sizes the literal arm is not attempted once its explicit expansion cap would be exceeded; those cases have no baseline completion time and no speedup ratio.

Grammar nodes	Expanded length bits	Median query and replay (ms)
132	109	0.492
516	434	1.767
2,052	1,736	7.652
8,196	6,940	40.762

Table 3: Nonuniform compressed inputs beyond the literal expansion allowance. Construction is included; no speedup ratio is assigned to an unrun baseline.

For the largest nonuniform case, the expanded length is at least 2^{6939} , while the timed compressed construction, query and replay have a median of about 0.041 seconds in this environment. The useful conclusion is that the implementation operates on the grammar rather than on all spelled letters. It is not a claim that a knot with 2^{6939} crossings has been recognized.

11.3 Whole source-bound braid demonstrations

The paired braid experiment contains 48 source braids: 36 known unknots obtained by conjugating stabilized circle braids, and 12 nontrivial torus knot examples with stabilizations. Both arms certify all 36 unknots and return **INCONCLUSIVE** on the other 12. Across the 48 cases, the sum of median query costs is approximately 4.163 milliseconds for the width arm and 4.446 milliseconds for the literal-Whitehead arm, a ratio of 1.068. Each query includes source construction and independent replay.

These small differences do not justify an automatic production dispatch rule. No statistical significance study was performed, the sample family is selected, and this is not the maintained recognizer. The experiment establishes a functioning source-bound certificate path and agreement of the two terminal choices, not an end-to-end speedup for the current repository. The much larger kernel ratios must not be substituted for this whole-query comparison.

11.4 Balanced contraction families

For a full binary tree of generator pairs, attach relations

$$(a^N b)^2 = 1, \quad (a^N b)^3 = 1, \quad N = 2^h,$$

at each merge, using the surviving left representative in the next level. The two powers force $a^N b = 1$ even without torsion-freeness, and the word $a^N b$ is primitive. Therefore each level eliminates the right member and the full presentation is $\mathbb{Z}$. This family is genuinely nonuniform in its relator spelling, but is synthetically arranged to admit the desired contractions.

Initial rank	Initial nodes	Rounds	Final nodes	Run time (ms)
4	78	2	139	1.32
16	318	4	273	8.18
64	1278	6	407	103.11
256	5118	8	541	215.05

Table 4: Single-run conditional contraction experiments on the explicit balanced presentation family, with $h = 32$. Initial-family construction is outside these contraction timings.

The records include the nodes, largest length-bit bound, generator count and chosen-pair count at each round. The observed success is a test of the implementation on a controlled schedule. It is neither evidence that all Wirtinger presentations admit such schedules nor a tightness experiment for the worst-case bound $S_0 + O(r_0 B_0 2^d)$.

12 Research questions with concrete success criteria

The following questions are proposed in an order that separates integration work from the universal structural bottleneck. A useful answer may be a positive theorem, a counterfamily, or a measured negative result.

1. Production frequency and dispatch. At which actual maintained search states do cyclically reduced two-supported primitive-power relators occur? Add read-only telemetry first, then a

versioned public terminal verifier. A meaningful performance result must compare fresh whole queries, include normalization and replay, and retain cases with no terminal activation. The present kernel benchmark does not answer this question.

2. Normalization without repeated polynomial blow-up. Can the eligibility of a raw compressed relator be tested through a persistent normal form with a global, additive grammar budget? A sufficient theorem would replace repeated $S \mapsto S^c$ bounds by $S_{j+1} \leq S_j + \text{poly}(r, B_j)$, or by another recurrence whose d -fold composition remains $\text{poly}(n)2^{O(d)}$. Such a theorem would enlarge the applicability of the round-depth result substantially.

3. Bounded-depth current-relator schedules. For a quantitatively specified class of planar Wirtinger presentations, do current primitive-power relators suffice for $O((\log n)^2)$ successful rounds after a controlled preprocessing stage? Both an efficient producer and a rank-reduction guarantee are required. A counterexample should separate failure of a specific greedy schedule from failure of every allowed schedule.

4. Canonical target proof length. At a certified rank-two interface, what upper bounds hold for the shortest normal-derivation program proving the signed Christoffel target of Theorem 9.1? An $O(\log n)$ bound together with a polynomial-size interface would yield the explicit quasi-polynomial positive search in Proposition 9.3. Polynomial witness size alone would not give that search bound.

5. Safe parallel scheduling. Can a greedy matching be replaced by a canonical selection rule with an exchange theorem ensuring that later contractions are not obstructed? Alternatively, bound the number of essential branch choices rather than all matchings. This would address the difference between a cheaply checked good schedule and a cheaply found one.

6. Primitive roots versus primitive relators in practice. Does torsion-free root elimination activate on residual knot presentations where ordinary primitive-relator elimination does not? Compare genuine powers, nonuniform powers, and redundant conjugate powers separately. Synthetic examples demonstrate possibility, not prevalence.

7. Larger free factors. Can a useful higher-rank primitive or free-factor certificate be evaluated by a similarly small exact summary? A one-dimensional exponent test cannot simply be assumed to classify primitive conjugacy classes beyond rank two. A practical intermediate goal is a certified, bounded-cost change of basis that exposes a two-generator-supported relator without allowing uncontrolled word growth.

8. Cooperation with meridian-preserving terminals. The current two-meridian terminal and the present arbitrary-basis terminal have different provenance requirements. Can a shared presentation pipeline retain both kinds of evidence and dispatch without reconstructing all of it repeatedly? A basis change is allowed here but may destroy the certified meridian hypothesis needed by another terminal. The evidence types must make that distinction explicit.

9. Complete restricted negative terminals. Integrate a verified deficiency-one reduction and the one-relator criterion of Theorem 7.1. Measure how often it produces a decisive negative answer before expensive homological work. An empty or malformed presentation endpoint must not be confused with an ordinary nontrivial-knot certificate.

10. Formal verification of the smallest trusted kernel. A suitable first formalization target is the prefix-profile monoid, the residue lower bound, the deterministic equality case, and the DAG induction. Next formalize the integer certificate checker and the primitive projection using an explicit free basis. Classical knot-group torsion-freeness and the Loop Theorem remain separate imported mathematical dependencies. No Lean artifact is claimed in this delivery.

11. Resource-accounted normal-consequence compression. Can derivations use long conjugators represented by small circuits without making their enumeration cost exponential in their expanded spelling? Checking a supplied compressed conjugator is easy relative to searching for one. The branching language, its encoding length and the effect on normalization must all be specified before a stronger search bound is stated.

12. The global recognition obligation. The central remaining question is to construct, on *every* input in a clearly stated class, either a controlled presentation/derivation reaching one of these exact terminals or a comparably bounded independent obstruction. For all-knot quasi-polynomial recognition, that class must include every classical knot diagram. An unbounded positive search, an existential balanced schedule and a fast terminal do not jointly establish the missing uniform producer.

13 Conclusion

The local improvement is an exact, independently replayable integer kernel: cyclically reduced rank-two primitive powers are recognized by minimum prefix-height width in quadratic worst-case grammar time. For a certified knot group, these words justify root elimination and a positive unknot terminal without a meridian assumption. For disjoint two-generator supports, the projection $a \mapsto t^v$, $b \mapsto t^{-u}$ leads to a controlled normalization-free composition whose cost is singly exponential in the number of contraction rounds, not in the expanded word length.

This supplies a precise quasi-polynomial route when a successful schedule has $O((\log n)^2)$ rounds and the input-to-presentation producer is also controlled. The delivery does not establish those hypotheses universally. The most consequential next theorem is therefore not another local speedup, but a costed structural or proof-search result that brings actual knot presentations to these terminals. The accompanying source, tests, raw measurements and integration contract make that distinction reproducible.

A Reference pseudocode

The following specifies the mathematical kernel. The code adds format, resource and cancellation validation.

```

primitive_power(root, binary_DAG):
    meta = bottom_up_counts_endpoints_reduction_flags(binary_DAG)
    require root is freely and cyclically reduced
    if length(root) == 0: return NOT_PRIMITIVE_POWER
    if either generator occurs with both signs:
        return NOT_PRIMITIVE_POWER
    U, V = signed_exponent_sums(root, meta)
    d = gcd(abs(U), abs(V))
    u, v = U // d, V // d
    weights = {a:v, inverse(a):-v, b:-u, inverse(b):u}
    profile[empty] = (0, 0, 0)
    for node in topological_order:
        if node is a letter of weight w:
            profile[node] = (w, min(0,w), max(0,w))
        else node = concat(X,Y):
            t,m,M = profile[X]; s,n,N = profile[Y]
            profile[node] = (t+s, min(m,t+n), max(M,t+N))
    total, low, high = profile[root]
    if high-low != abs(u)+abs(v)-1:
        return NOT_PRIMITIVE_POWER
    return CERTIFICATE(root, u, v, d, high-low)

```

No root string, Christoffel word or Nielsen basis is built in this query. Such constructions are useful for interpretation and other endpoints, not required for the arithmetic decision itself.

B Claim and implementation ledger

Claim	Mathematical status	Software status
Minimum-width primitive-power criterion	Proved here, using classical rank-two classification for the converse	Implemented and independently replayed
Quadratic cyclically reduced grammar query	Proved in Theorem 4.2	Exact integer implementation; explicit caps
General compressed normalization	Imported polynomial-time theorem	Not implemented by the standalone kernel
Torsion-free primitive-root elimination	Proved in Theorem 6.3 and preceding root-elimination statement	Local certificate kernel; full provance required
Positive arbitrary-basis rank-two knot terminal	Proved in Corollary 6.4	Source-bound literal braid demonstration; no maintained production hook
Complete one-relator terminal	Proved using Magnus's theorem	Not installed as a negative knot-verdict path
Parallel depth bound	Proved for the stated raw-circuit rounds	Conditional prototype; no independent whole-run checker

Claim	Mathematical status	Software status
Bounded normal-derivation enumeration	Proved in Proposition 9.3	Not implemented
Universal quasi-polynomial recognizer	Not established by this report	No claim of implementation

C Reproduction and repository provenance

The top-level `README.md` gives runnable commands. Tests require only Python’s standard library. Building the article additionally requires a LaTeX installation with the packages listed in its preamble. The timing report is generated by `code/benchmark.py`; the table files are derived by `code/make_tables.py`. Seeds and raw samples are retained, so an independent run need not reproduce exactly the same elapsed times to check the same mathematical outputs.

The repository head observed during the review, dated 8 October 2026, was

c4c23b61d844b465b9ee30d0cebb02af82050306.

The audit used individual connector-fetched blobs, with exact hashes and read ranges recorded in `data/provenance.json`. It was not a checkout of the entire repository. In particular, only the first 165 lines of the compressed-word module were inspected for the adapter schema, while the compressed search and the cited synthesis sections were read as recorded. No full-repository regression result is asserted. No remote repository files were changed.

The archive includes only this report’s original text and research code, the generated PDF, local data and supporting integration notes. It does not redistribute the cited external papers or font files. Checksums establish artifact integrity, not mathematical correctness.

References

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